

LAB 4: Matrix Arrays & Regular Expressions

- **Syllabus Alignment:** Unit 2: Strings and Arrays (6 LHs) — Arrays (Definition, Indexed, Associative, Multidimensional), `foreach` loop, Array Functions, Regular Expressions
- **Lab Objectives:** To handle indexed collections, key-value maps, and nested tables, while using regular expressions to validate text patterns.
- **Lab Questions:**
 1. **Indexed Arrays:** Create a numerically indexed array named `$months` containing calendar month strings. Traverse and display the entries using a traditional `for` loop bounded by the `count()` function.
 2. **Associative Maps:** Use an associative array structure to map month names to their exact number of days. Iterate over this mapping using a `foreach` loop to print messages like *"January has 31 days"*.
 3. **Multidimensional Arrays to HTML Tables:** Build a multi-dimensional array containing personal details (Name, Age, Gender) for four individuals. Write a loop that renders this nested array into a clean, well-formatted HTML table containing columns for *S.No.*, *Name*, *Age*, and *Gender*.
 4. **Regular Expression Engine:** Build a backend pattern matching block that utilizes `preg_match()` expressions to validate fields from user submissions (e.g., ensuring a name field contains only pure alphabetic characters, or validating a 10-digit mobile phone number starting with 97 or 98).